FSODS: A Lightweight Metalearning Method for Few-Shot Object Detection on SAR Images

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***Abstract*— At present, few-shot object detection research in the field of optical remote sensing images has been conducted, but few-shot object detection in the field of synthetic aperture radar (SAR) images has rarely been explored. To this end, this article proposes a lightweight metalearning-based SAR image few-shot object detection method, which improves the accuracy and speed of SAR image few-shot object detection from a more balanced perspective. First, we introduce the latest FSODM method in optical remote sensing as a benchmark framework. Second, a lightweight metafeature extractor named DarknetS is designed to enhance the feature representation of SAR images and improve detection timeliness. Furthermore, we build a new aggregation module called AggregationS, which encodes support features and query features into the same feature subspace via a novel transformer encoder. This module design can better extract the correlation and saliency between different classes in the support set, improve the detection accuracy of the query set, and enhance the detection generalization performance of new classes. Finally, we built several real-world SAR image few-shot object detection datasets to verify the effectiveness of the method. Experimental results show that FSODS can achieve a better object detection performance compared to the baseline model under the condition that only a small amount of labeled data is required for new classes of SAR image objects.**

***Index Terms*— Few-shot, lightweight, object detection, synthetic aperture radar (SAR), transformer.**

I. Introduction

SYNTHETIC aperture radar (SAR) is an indispensable and important monitoring tool in the field of remote sensing. It is an active microwave imaging sensor and the main way to obtain SAR images using all-weather, all-day imaging, and reconnaissance capabilities. It can provide high-resolution images independent of weather and lighting conditions, and has been widely used in many industries [1]–[5]. With the rapid development of airborne SAR and spaceborne SAR for decades, much research has been carried out in SAR object detection. The detection of multiscale SAR objects in complex scenarios is one of its main tasks and remains a pressing challenge for researchers.

In the recent decades of artificial intelligence development, object detection has been a key research task for researchers, and a series of research results have been achieved. In the field of SAR image object detection, a large number of models and methods have been developed by researchers to detect objects in SAR images (e.g., tanks, ships, aircraft, and bridges). The approaches based on contrast information [6]–[8], geometric and texture features [9], [10], and statistical analysis [11], [12] are basically included in the traditional object detection methods for SAR images. The constant false-alarm rate (CFAR) method [13] is a common object detection technology in the proposed object detection algorithm for SAR images. To distinguish an object pixel from the background, the CFAR method first calculates the self-adaptation threshold according to the given false alarm rate and the statistical distribution of background clutter and then compares the pixel intensity with the calculated threshold. The performance of this method is greatly influenced by the goodness of the parameter estimation of the model used and the performance of the statistical model of sea clutter. Therefore, many improvements have been proposed, mainly from these two aspects. Both the symmetric alpha-stable distribution [14] and generalized gamma distribution [15] are based on a nonuniform distribution to fit the changing sea state because of the nonuniformity of clutter and various clutter models. However, the parameter estimation becomes difficult and time-consuming as the complexity of the model increases. A practical application is considered by Gao *et al.* [16] in an attempt to achieve a certain balance between speed and accuracy estimation. Wan *et al.* [17] improved on the basis of YOLOX [18], made a lightweight design for its backbone, and used a new attention and a new target detection head, and the accuracy and speed reached a certain balance on the SAR Ship Detection Dataset (SSDD). Xia *et al.* [19] combined CNNs with a transformer to

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extract more abundant global information from SAR images, and the accuracy of the SSDD dataset reached a relatively high level. Our research group participated in the construction of a new SAR multiclass target detection dataset SMCDD, and Xia *et al.* [19] verified the effectiveness of his method.

In addition, the high speed of machine learning and GPU computing power have led to significant breakthroughs in convolutional neural networks (CNNs) for object detection. Different from the traditional methods using artificially designed features, the method based on machine learning [20] has stronger robust feature extraction and scene classification capabilities. Therefore, from the extracted data, the deep neural network can easily and automatically learn its feature representation. A large number of CNN-based object detection methods have been investigated to solve problems in their respective fields, not only in the field of optical images but also in remote sensing SAR images. At present, there are basically two kinds of object detectors in object detection methods: anchor-based detectors and anchor-free detectors. YOLO [21]–[23], SSD [24], Faster R-CNN [25], and so on are among the very classical object detectors, which have been tested in many engineering and experiments, and have been widely applied to various projects. In the field of SAR image object detection, most of the current research is based on the mainstream framework of computer vision.

Although a variety of deep learning models and methods have been proposed in the field of object detection in recent decades, all of these methods first require large and diverse datasets to train deep neural network models, especially in the military domain, where the reality may not allow us to be able to collect as much new SAR data with extensive manual annotations, such as enemy airplane tanks. Second, if these methods retrain their parameters on the collected new datasets, it will take a lot of time. If a small number of annotated samples are taken from them to train the network model to fit the realistic situation, it is easy to overfit, and the generalization ability of the model is greatly reduced. Therefore, we should comply with the setting in the military field, that is, only a small number of enemy SAR target images can be obtained from the enemy. We need a special learning mechanism to learn certain feature knowledge from a small number of new class samples.

Therefore, considering the sample factor, few-shot learning has emerged to conform to realistic settings and has been developed and studied to some extent and applied in many computer vision-related tasks, such as few-shot image segmentation [26]–[28], few-shot object classification [29]–[31], and few-shot object detection [32]–[35]. The obvious goal of few-shot learning is that the model learns transferable knowledge from a large number of labeled base classes datasets that are easily accessible and then recognizes images of the new class from a dataset with only a few labeled samples needed for training; the knowledge learned must be well extended to the new class. However, in the field of few-shot learning, many researchers use it for image classification tasks, and it is only in recent years that few-shot learning has been applied to optical images for object detection tasks. There are few applications of few-shot object detection to remote sensing images, especially in the field of SAR image target detection, which is more in line with realistic settings and military needs. Basically, the classification task is performed on the MSTAR dataset [36]. Compared to optical images, SAR images have longer imaging wavelengths and more complex imaging mechanisms, and the imaging results are more difficult to interpret intuitively. Moreover, the objects are discontinuous in SAR images, i.e., they are composed of multiple discrete irregular scattering center bright spots, and at the same time, the objects in SAR images are not uniformly distributed. Their sizes may be very different, their spatial resolutions may not be consistent, and there is great sparsity, which makes it more difficult for few-shot object detection to effectively detect objects from a new class with only a few new class samples of SAR images trained with labels.

Therefore, this article proposes a method for few-shot object detection on SAR images that is based on lightweight metalearning to increase the accuracy and speed of few-shot object detection from a more balanced perspective. In consideration of the possibility of using the few-shot set in the modern military field, the method that we proposed in this article is able to learn more knowledge from the easily obtained SAR dataset with a large number of labeled base classes and can apply this knowledge to a new class that is difficult to obtain and has only a small number of well-labeled samples for object detection.

The basic idea of our SAR image few-shot object method is given in Fig. 1. Our method consists of two phases: the base class training phase and the new class fine-tuning phase. In the training phase, our model is trained on an easily accessible base class dataset with a large number of annotations, and then, in the new class dataset fine-tuning phase, the metaknowledge learned from the base classes dataset can be applied to the new class dataset with only a small number of annotations, and only a few fine-tuning training sessions are needed for its fast convergence and identification of new class dataset to achieve a certain performance.

The primary contributions of this article are summarized as follows.

1) First, for few-shot object detection on SAR images, we innovatively design a few-shot object detection method based on lightweight metalearning. Inspired by the latest framework for few-shot object detection in the field of optical remote sensing, we introduce FSODM [37] as the benchmark network; furthermore, to meet the edge-end deployment for practical engineering applications and address the unique characteristics of SAR images, we design a new lightweight metafeature extractor, called DarknetS, which maintains good performance despite the large reduction in the number of parameters and FLOPs compared to the original model.
2) Second, to enhance the salient feature representation of SAR few-shot objects, we construct a new aggregation module of support features and query features called AggregationS. This module recodes the learned set of support features of each class into the same space by a transformer encoder and then performs a channel product with query features encoded by a transformer encoder, which can better learn the salient semantic

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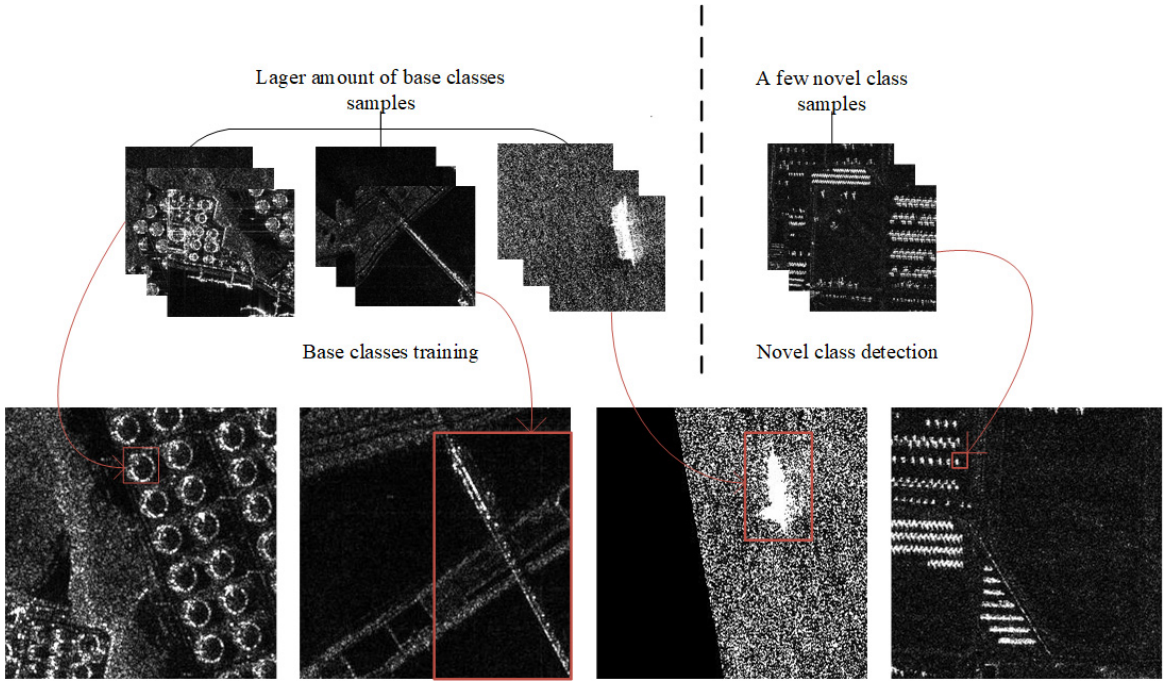


Fig. 1. Interpretation of few-shot object detection on SAR images. A large number of annotation samples from base classes are trained using our model and use a small number of annotation samples to detect objects from the new class.

features to improve the accuracy of the query set when making queries in the support set, reduce misclassification, allow better generalization of the knowledge learned in a large base classes dataset, and enhance the detection and recognition rate of the new class.

3) Furthermore, to effectively alleviate the problem of low detection efficiency caused by sample imbalance and postprocessing anchor frame suppression, we design a focal loss [38] that is more adapted to sample imbalance training and Diou-NMS [39] at the postprocessing end to improve the accuracy of few-shot object detection.
4) Furthermore, to verify the effectiveness of our proposed method on a wider range of datasets, we construct three real-world SAR image few-shot object detection benchmark datasets. The source data come from the images of typical global regions collected by China's first commercial remote sensing SAR small satellite-HISEA-1, which was developed and launched by our research team. In the future, we will release the above SAR image few-shot multiclass object detection dataset.

II. Related Work

A. General Object Detection in Computer Vision

The detection task has become a mainstream process in the field of computer vision in recent decades. It has a long history and has had an upsurge in usage with the rise of deep learning. Approximately, a decade ago, most research works emphasized the use of the sliding window method for object detection studies. With the increasing computational power of GPUs, deep CNN methods in deep learning are being widely used. Currently, deep CNN-based approaches are basically divided into two classes based on the presence or absence of anchor frames: one-stage algorithms and two-stage proposal anchor-based algorithms. The single-stage method removes the process of region suggestion generation, directly performs object detection on the input image, and regresses the target coordinate information and category information directly at the prediction end to finally obtain the detection results. YOLO [21]–[23] is a very classical single-stage algorithm and is one of the most successful single-stage algorithms that are widely used. The YOLO model will further divide the input image into the form of grid cells, each of which has its own task; they will be responsible for the detection of a fixed number and size of objects. Compared to two-stage object detection algorithms, YOLO may not perform, as well as two-stage algorithms, but the detection rate is much faster than two-stage expressions. Because the YOLO model combines and achieves a good balance of detection speed and accuracy, YOLO may be more suitable for engineering applications than the two-stage algorithm. Among the two-stage algorithms based on proposal anchors, the first method to propose a solution to the problem of object localization using the proposal anchor-based approach was R-CNN [40]. The authors use a CNN-based approach to extract features instead of the traditional handcrafted process with significant performance improvements. Fast R-CNN [41] is an upgraded version of R-CNN that replaces the method of serial feature extraction in R-CNN, directly uses a neural network for feature extraction of the original input image, maps the features of each region of interest (ROI) into a fixed-length vector by the proposed ROI pooling layer, and greatly improves the speed and detection efficiency of the algorithm. Faster R-CNN [25] is another variant of R-CNN, which generates region proposals by the

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proposed region proposal network (RPN) on the basis of R-CNN, and the front- and back-end networks achieve weight sharing, thus enabling end-to-end training. It becomes the most frequently used target detection algorithm in the later dual phase, and most of the work has been built based on Faster R-CNN. Examples include Mask R-CNN [42].

Compared with common optical images, SAR image object detection is also a field that has just been developed in recent years, and its development is not as rapid as that of common optical image object detection because, first, SAR image resources are scarce, and second, unlike the imaging mechanism of optical images, the imaging mechanism of SAR images is more complex, so it is more difficult to identify and detect objects. Object detection on SAR images is basically developed on the basis of optical image object detection, which has a certain combination of knowledge information specific to SAR images. However, in a more realistic setting, effective object detection using only a few training samples on a new class of SAR images is still largely missing in practice and is more often performed for classification tasks.

B. Object Detection in SARIs

Currently, SAR image object detection is generally performed on ships in SAR images because the publicly available SAR datasets are basically SAR ship targets, such as SSDD [43] and AIR-SARShip [44]. The traditional ship detection algorithms are basically improved from CFAR [13]. Specifically, to achieve the fusion of intensity and spatial information, an algorithm named intensity-space domain CFAR was proposed by Wang and Liu [45]. A similar idea was later developed by Pappas *et al.* [46] using superpixels to distinguish the object from the alert band of CFAR. However, these algorithms may not be able to handle such a complex scene because ships in SAR images are often in a complex marine context. Later, to be able to select the candidate regions to be detected and obtain the threshold values for detecting the presence or absence of ships, Huo *et al.* [47] adopted the maximum-stable extreme region algorithm. To achieve the separation of the ship from the background, Shi *et al.* [48] used a directed gradient histogram for feature extraction. However, all of these methods share the common disadvantage that small targets cannot be effectively detected. Due to the continuous updating of algorithms and the improvement of the computing power of the hardware, deep learning technology has been increasingly widely used in recent decades. A large number of researchers have begun to apply different deep CNNs to carry out object detection research on SAR images. CNNs have more obvious advantages over traditional methods and can easily achieve end-to-end object detection tasks without designing complex feature extraction experiments. Chen *et al.* [49] proposed an effective detection network algorithm that enables the detection of multiscale SAR ships in complex scenarios, where the attention mechanism plays an important role. To enable adaptive multiscale ship detection, Cui *et al.* [50] designed a dense attention pyramid network (DAPN) by introducing a convolutional attention module, but the complex inland background could not be detected well either. Therefore, Jiao *et al.* [51] improved the algorithm from Faster R-CNN by using a densely connected multiscale CNN to better handle the detection of SAR ships in multiple scenes. To solve the problem of feature misalignment and prediction of rotating bounding boxes, a fine single-stage detector with feature refinement (R3Det) was proposed by Yang *et al.* [52]. To find a reasonable balance between speed and accuracy, Fu *et al.* [26] fused the attention-guided balance pyramid with the refined detection head to implement ship detection based on the anchor-free method. Wan *et al.* [17] further improved the YOLOX [18] model based on this idea and made a lightweight design for its backbone. Finally, they realized multiscale anchor free frame detection on the SSDD dataset and achieved good performance. Recently, Xia *et al.* [19] improved the Swin transformer, applied it in the field of SAR image target detection, and achieved good performance, and our research group participated in the construction and cleaning of a new SAR multiclass target detection dataset SMCDD [19]. However, these object detection methods for SAR images are carried out under the condition of sufficient data; this article focuses on the solving problem of few-shot object detection on SAR images under the condition of few training samples. In general, in the field of machine learning, we usually consider two processes when designing object detectors: manual feature extraction and classification. Compared with other traditional methods, the method based on machine learning shows a more powerful generalization ability. The traditional method based on machine learning uses manual features, while the method based on deep learning uses CNN to learn robust features automatically from the input image. Therefore, in recent years, the method used in deep learning has been further studied in SARIs object detection work.

C. Few-Shot Object Detection

The term metalearning has become very popular in recent years, and its derivative, few-shot learning, has become a new research hot topic in recent years. Few-shot learning aims to simulate realistic scene settings to obtain certain recognition performance on small numbers of samples by training the model with only a small number of training samples to achieve object recognition (such as classification, detection, and segmentation) in few-shot situations. Early metric learning methods, such as Siamese neural networks [53], matching networks [54], and prototype networks [55], also belong to early few-shot learning, which mainly focuses on classification problems. However, if the task to be done is few-shot object detection, it often does not show good performance because target detection is a more difficult task than classification, which requires not only the implementation of the classification task but also the analysis of the target's location information. Currently, there are two main approaches to do few-shot object detection: fine-tuning-based methods and metalearning methods, both of which divide the training dataset into two parts: a large number of base classes data with labels and a new class of data with only a few or tens of labels. The metalearning approach designs a new framework that consists of metatraining and original testing, where metaknowledge is

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trained on the base classes dataset and transferred to the new class. Meta R-CNN [56] proposes a predictive head reshaping network (RPN) for ROI metalearning to predict the new class by inferring the class attention vector. Cao *et al.* [57] proposed a two-stage training method to solve the small sample detection task of plateau wildlife images. Kaul *et al.* [58] proposed enriching the training samples in a semisupervised manner by training a good initial model to label additional unannotated samples and obtained a fairly good performance. However, it does not match the realistic military domain with few-shot settings since it is assumed that we cannot obtain many samples from the enemy. FSFR [35] designed a DarkNet-19 backbone-based YOLOv2 [22] metalearner, which introduces a metafeature learner and a feature reweighting module. The metafeature learner DarkNet-19 network is used to extract metafeatures and then is weighted with a set of reweighting vectors of each class generated by the reweighting module from several support samples. Finally, the detection result generated by the bounding box prediction module is obtained by using the reweighted element features. The FSODM extends it to three scales, namely, YOLOv3, whose backbone is based on DarkNet-53, and applies it to remotely sensed images. However, the Darknet-53 network in FSODM is deeper, and the number of parameters is too large, which tends to cause an overfitting phenomenon for training samples and is not conducive to porting to embedded devices. Unlike FSODM, we redesign the metafeature extractor to be lightweight from the perspective of lightweighting and practical applications, and improve the reweighting method of a single channel product so that the number of parameters and FLOPs is greatly reduced. The performance can still exceed FSODM in the few-shot case.

III. Methods

A. Methodology Overview

How to accurately describe and set up the problem of few-shot object detection is the premise of studying few-shot object detection. The ability of the few-shot object detection model that we designed to learn metaknowledge from the base classes dataset with a large number of label annotations, to efficiently migrate it to only new class datasets with few label annotations, and, finally, to detect new class objects quickly and efficiently is among the most important difficulties to be solved.

This problem of few-shot object detection is structured for application to a real-world military setting because, in the military field, we may only get a few SAR images of enemy military objects, and we need to rely on these few images to be able to identify the enemy military objects more accurately the next time when we encounter them to make accurate judgments. Therefore, we need to design a model for few-shot object detection on SAR images. It needs to be able to open the next new class efficiently without training a new class dataset from scratch because this is a very time-consuming and low generalization ability. We need to consider a pretrained detection model with metaknowledge on the existing large-scale SAR object detection dataset (e.g., HISEA) using metalearning and then open the new class directly when it comes. Then, we can just turn on the new class directly when it comes.

The metalearning few-shot object detection method reconstructs our partitioned base classes training set $\mathbb{B}_i$ and a new class training set $\mathbb{N}_i$ into many subsets $\mathbb{E}_i$, which, in turn, consists of a set of query images $\mathbb{Q}_i$ from the same class set and a set of support images $\mathbb{S}_i$ with annotated labels and an image from each class of the same class set. Assuming that there are three classes in the base classes dataset and one class in the new class, training on the base classes dataset is the task of partitioning into sets of three samples each, and each support set consists of one labeled image from each class of the base classes objects. The support set is denoted as $\mathbb{S}_i = \{\mathbb{I}_k, \mathbb{M}_k\}\ k = 1, 2, 3$, where the input images of each class are denoted as $\mathbb{I}_k$, $\mathbb{I}_k \in \mathbb{R}^{h\times w\times 3}$, and $\mathbb{M}_k$ denotes the true labeling of the images. When new classes are added to train together, k is expanded to 4. The images in the support set are used by our model to learn metaknowledge, and then, the learned metaknowledge is applied to the query set images to perform object detection.

Fig. 2 shows the overall architecture of our model. Our model of few-shot object detection (FSODS) for SAR images first learns metaknowledge from a large base classes dataset with labeled data, so we first need our designed lightweight metafeature extractor module DarknetS to extract features from the input query image, which has three different scales, and then, the reweighting module processes the input support images with annotated labels for each class and outputs three sets of corresponding reweighted vectors, one vector for each class in each set. The extracted query features of the three scales and the three sets of corresponding reweighted vectors are recalibrated by our redesigned aggregation module AggregationS and, finally, output to the three prediction layers to obtain the final prediction results. By using the AggregationS module, we recode the features from samples in the query set and those in the support set into the same feature space and highlight the important metafeatures of each class and the differences of the features before each class, which are more effective in detecting the object information on the query image.

B. Lightweight Metafeature Extractor: DarknetS

We redesigned a lightweight metafeature extractor that we call DarknetS, which can extract three scales of query metafeatures from the input query image. Unlike FSODM [37], which uses DarkNet-53 as its metafeature extractor, we start from the perspective of practical engineering applications and lightweighting because the number of parameters extracted and FLOPs computed by DarkNet is too large for real practical applications to be ported to embedded devices, and the excessive number of network parameters can easily cause overfitting of training samples, especially in the setting of few-shot where the sample size is too disparate. Therefore, on the basis of DarkNet53, we redesigned it to be lighter, we called it DarknetS, and it was constructed based on MobileNetV3 and the characteristics of SAR images.

DarknetS first replaces all the 3×3 convolutions inside each block in DarkNet53 with the depthwise separable

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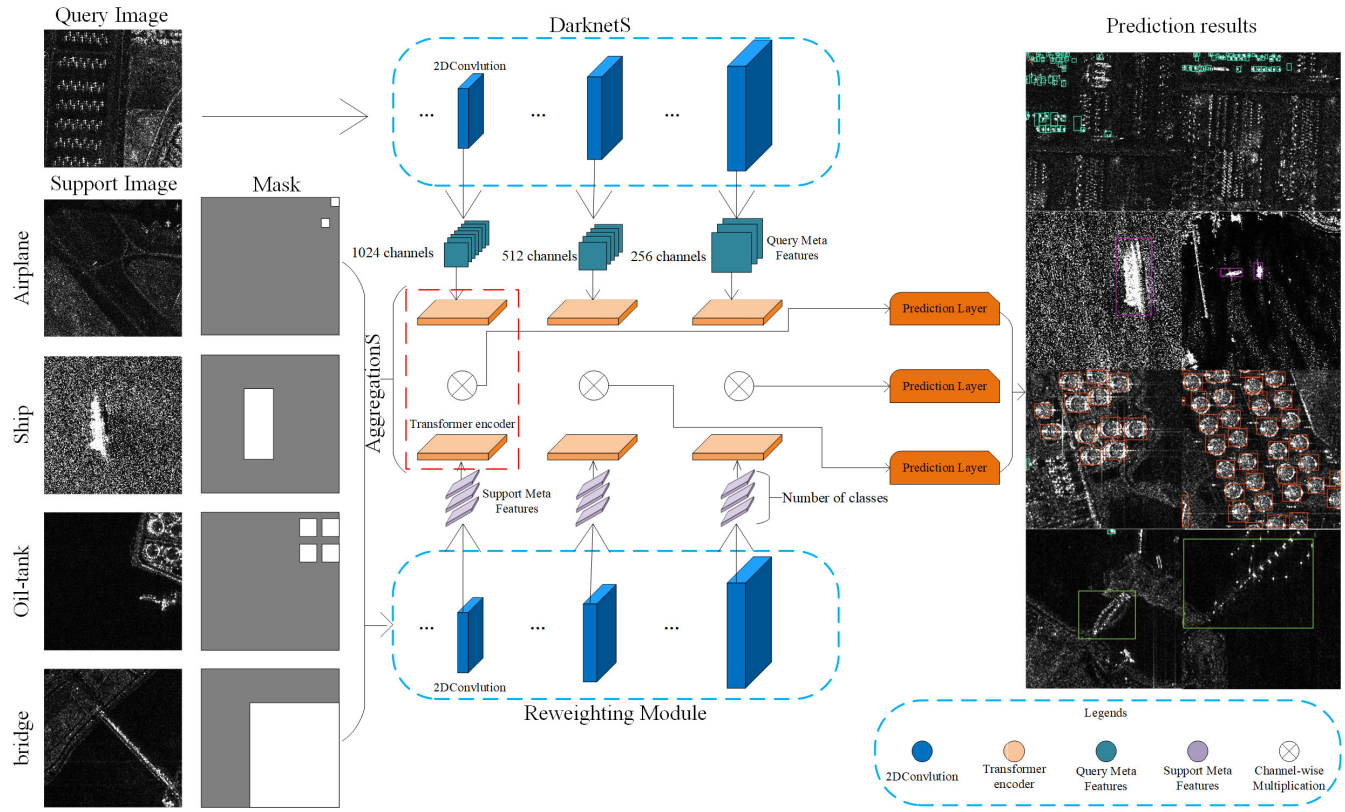


Fig. 2. Overall architecture of our proposed model for few-shot object detection on SAR images. Among them, DarknetS represents our designed lightweight metafeature extractor. AggregationS represents our proposed transformer-based metafeature aggregation module, which aggregates two sets of metafeatures.

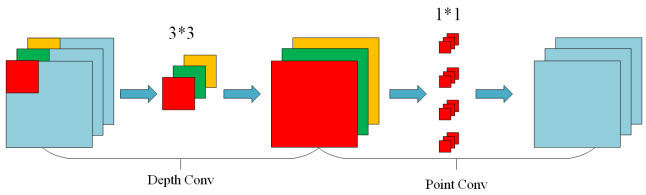


Fig. 3. Depthwise separable convolution.

convolutions used in MobileNetV3 [59]. Channel-separable convolution is the main feature of the MobileNet family and the main factor in its lightweight role. In Fig. 3, the channel-separable convolution is divided into two processes:

1) channel direction channel-separable convolution;
2) normal 1×1 convolution output with the specified number of channels.

Our DarknetS also introduces the squeeze-and-excitation (SE) [60] channel attention module used in MobileNetV3. The core idea of this module is to improve the quality of the network generated by displaying the interdependencies between the convolutional feature channels of the network. Specifically, we need to boost the useful features and suppress the noisy features that are distracting from the task at hand, so the importance of each feature channel needs to be automatically acquired through the learning of the network. Thus, through this module, the network can selectively emphasize the informative features in SAR images and suppress the less useful SAR image noise features to learn the global information. Compared with the conventional SE module, which is essentially the same as FC, here, it utilizes a 1×1 convolution to implement the FC operations and utilizes the less computationally intensive H-sigmoid instead of sigmoid operations

$$\text{H-sigmoid} = \frac{\text{ReLU6}(x+3)}{6}. \tag{1}$$

Because the SE structure consumes some time, the authors change the channel of the expansion layer to 1/4 of the original one in the structure containing SE so that the authors find that the accuracy is improved without increasing the time consumption. The proposed FSODS model also adds the improved SE module to each block after the depth convolution and before the point convolution. The specific operation flow of the SE module is shown in Fig. 4.

To reduce the number of operations and improve the performance, we also use the H-swish activation function instead of the swish function in the deep convolution in the structure of FSODM's deep separable convolution H-swish; it is expressed as follows:

$$\text{H-swish} = x \cdot \text{H-sigmoid}. \tag{2}$$

Finally, we change the structure of the approximate residual in DarkNet53's block to the inverted residual with a linear bottleneck (the inverted residual with a linear bottleneck) as in MobileNetV2 [61], i.e., it is first updimensioned using

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TABLE I
NETWORK STRUCTURE OF DARKNETS

Block Nums	Type	Filters	Size	Output
	Convolutional	8	3×3	256×256
	Convolutional	16	3×3/2	128×128
1×	Convolutional	64	1×1	
	Depth Conv	64	3×3	
	Point Conv	16	1×1	
	Residual			128×128
	Convolutional	32	3×3/2	64×64
2×	Convolutional	128	1×1	
	Depth Conv	128	3×3	
	Point Conv	32	1×1	
	Residual			64×64
	Convolutional	64	3×3/2	32×32
8×	Convolutional	256	1×1	
	Depth Conv	256	3×3	
	Point Conv	64	1×1	
	Residual			32×32
	Convolutional	128	3×3/2	16×16
8×	Convolutional	512	1×1	
	Depth Conv	512	3×3	
	Point Conv	128	1×1	
	Residual			16×16
	Convolutional	256	3×3/2	8×8
4×	Convolutional	1024	1×1	
	Depth Conv	1024	3×3	
	Point Conv	1024	1×1	
	Residual			8×8

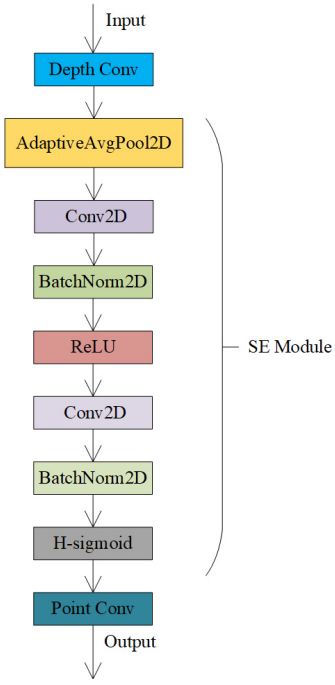


Fig. 4. Structure of the SE module.

a 1 × 1 convolution, and then, a depth-separable convolution is downdimensioned with residual edges. MobileNetV2 changes the bottleneck structure into a spindle type, that is, resnet is first reduced to 1/4 of the original and then enlarged, but MobileNetV2 is enlarged to six times the original and then reduced. Due to the limited computational resources, our FSODS is scaled up to four times the original and then scaled down. The overall DarknetS is shown in Table I.

C. Feature Aggregation Module: AggregationS

The transformer-based [62] metafeature aggregation module is used to aggregate two groups of metafeatures. One group is the support metafeatures extracted from the support images with labels by the reweighting module of the lightweight CNN. Each group of reweighting vectors is mapped from each support image, and each class corresponds to one reweighting vector. The other group is the query metafeatures that are extracted from the query images by our DarknetS, and they are reweighted by the reweighting vectors to highlight the useful information of each query feature, which is more beneficial for targeting queries in the query images.

Suppose that we have N samples of support images of the object class, and these N support images and their labels will be input to the feature reweighting module. Each input set of support images is composed of one support image I_j and label M_j drawn randomly from these N classes in the support set. After passing through the feature reweighting

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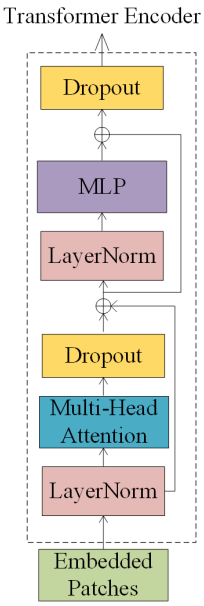


Fig. 5. Structure of the transformer encoder module.

module, it is mapped into a set of feature vectors, one for each class, denoted as $V_{ij} = M(I_j, M_j)$, and then, this set of feature vectors is reencoded by the transformer encoder module that we designed in Fig. 5 to obtain V'_{ij}, denoted as $V'_{ij} = E(V_{ij})$. The three sets of query image metafeatures F_i extracted by DarknetS are also encoded by the transformer encoder module and are denoted as $F'_i = E(F_i)$. Since the dimension of each reweighting vector is the same as the dimension of the corresponding metafeatures, we then multiply them by the channel to obtain the final set of three feature mappings to be output to the prediction layer for prediction

$$F_{ij} = F'_i \otimes V'_{ij}, \quad i = 1, 2, 3 \text{ and } j = 1, 2, \ldots, N. \tag{3}$$

As shown in Fig. 2, the transformer encoder and channel product form the AggregationS module of our FSODS. The transformer encoder block captures global information and rich contextual information. Each transformer encoder block is composed of two sublayers of a multihead attention layer and a fully connected layer [Multilayer Perceptron (MLP)], and a residual connection is used between each sublayer. The transformer encoder block not only exploits the self-attention mechanism to tap the feature representation potential but also increases the ability to capture different local information. Therefore, the AggregationS module we designed is used to highlight the object feature information from the SAR image, attenuate the noise information from the image background, and capture the interclass association between different classes, which greatly reduces the misclassification of similar classes and enhances the generalization of knowledge to a new class.

Fig. 6 shows the specific encoding process of the AggregationS module that we designed. The extracted query metafeatures and a group of support metafeatures are jointly input into the AggregationS module. The multihead attention reencodes these metafeatures through the three values of K, Q, and V so that the query features and support metafeatures of the same class can be encoded into similar feature distribution. The metasearch features and support metafeatures of different classes can be encoded into feature distributions with certain differences so that their intraclass correlations and interclass differences can be captured, so as to reduce misclassification and enhance model generalization, so that the query images can be queried correctly on the support images.

D. Focal Loss

The biggest problem of few-shot object detection is usually the imbalance of the sample, the number of base classes and a new class is often very different, so the classical detection algorithm is often trained to overfit the model, and the generalization ability is very weak, so we want to alleviate this problem of sample imbalance and weak generalization ability. We replace the original FSODM cross-loss entropy as a function of classification loss with the focal loss function [38]. The focal loss is specially designed for the one-stage detection algorithm, which reduces the weight of a loss that can easily distinguish negative cases so that the network will not be biased by a large number of negative samples. The equation for the focal loss function is shown as follows:

$$\mathrm{FL(pt)} = -(1 - p_t)^{\gamma} \log(p_t). \tag{4}$$

Among them

$$p_t = \begin{cases} p, & \text{if } y = 1 \\ 1 - p, & \text{otherwise} \end{cases} \tag{5}$$

where γ is a constant, and when it is zero, FL is the same as the ordinary cross-entropy loss function.

E. DIoU-NMS

Our FSODM replaces the conventional NMS with DIoU-NMS [39] for more accurate prediction at the postprocessing end because, in conventional NMS, the IoU metric is commonly used to suppress redundant detection boxes, where the overlapping region is the only factor that often produces false suppression for occlusion cases. The IoU equation is expressed as follows:

$$\mathrm{IoU} = \frac{\left|B \bigcap B^{gt}\right|}{\left|B \bigcup B^{gt}\right|}. \tag{6}$$

Among them

$$B^{gt} = \left(x^{gt}, y^{gt}, w^{gt}, h^{gt}\right) \tag{7}$$

represents the real frame.

Because the distance between the overlapping area and the center point between the two boxes should be considered in the suppression criterion, and DIoU considers the key to these two problems at the same time, DIoU-NMS takes DIoU as the NMS criterion. If the highest score is predicted, the formula of DIoU-NMS can be expressed as follows:

$$s_i = \begin{cases} s_i, & \mathrm{IoU} - \mathcal{R}_{\mathrm{DIoU}}(M, B_i) < \varepsilon \\ 0, & \mathrm{IoU} - \mathcal{R}_{\mathrm{DIoU}}(M, B_i) \geq \varepsilon \end{cases} \tag{8}$$

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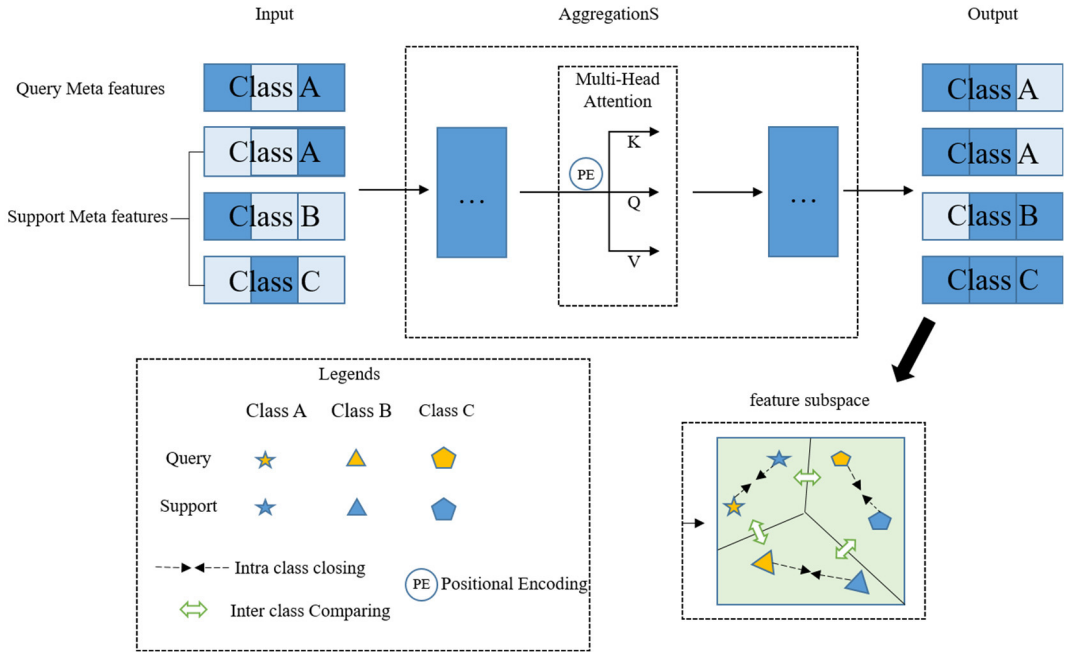


Fig. 6. Detailed process of the AggregationS module encoding.

where box B_i is removed by considering both the IoU and the distance between the centroids of the two boxes, ε is the NMS threshold, and s_i is the classification score. The major difference between DIoU-NMS and NMS is that DIoU-NMS suggests that two boxes with distant centroids may be located on different objects and should not be removed.

F. Training and Inference

To satisfy training in the few-shot object detection setting, the training set is repartitioned into two subsets, the support set (S) and the query set (Q), during our FSODS training, where the support set is divided by the target class in the training set. As mentioned before, if there are N classes of objects in the training set, we divide them into N groups, $i = 1, 2, 3, \ldots, N$. Each support image is input with its own true label. The support set is represented by the following equation:

$$S = \left\{ \left(I_1, M_{I_1}\right), \left(I_2, M_{I_2}\right), \ldots, \left(I_N, M_{I_N}\right) \right\}. \tag{9}$$

A group of query images and their annotations (A) form a query set

$$Q = \{(I, A)\}. \tag{10}$$

The training set is divided into many such sets, each of which can be expressed as

$$E_j = Q_j \cup S_j. \tag{11}$$

The query image is input to the feature extractor shown in Fig. 2, and the support image is input to the reweighting module shown in Fig. 2.

In the few-shot problem setting, we first select a certain class from the given dataset as a new class, so the dataset will be artificially assumed to be divided into base classes and new classes. The samples of the base classes are not cut, and the original state with a large number of labels is maintained. The base classes dataset is trained on our model to produce a basic model with better performance, and then, the new class is opened to start a new training task on the new class. The new class is a small number of randomly selected samples with labels from the original data, which is used as a realistic set of a new class, and only so many samples can be obtained.

Therefore, our training process consists of two parts: first, we train on the base classes dataset with a large amount of data, which will take longer, and we do not need to train on the base classes dataset again in the subsequent training; then, we open the new class with the trained model based on the first part, and we only need to train a few epochs to achieve better performance.

TABLE II
BASIC INFORMATION OF THREE EXPERIMENTAL DATASETS

	SMCDD-FS	SFSD	TFSD
public or not	public	nonpublic	nonpublic
train: valid	7:3	8:2	2:1
number of class	4	2	3
new class	airplane	ship	tank

IV. EXPERIMENTS

In the experimental section, to evaluate the performance of our FSODS model, we conducted extensive experiments on a public benchmark SAR object detection dataset and other nonpublic SAR target detection datasets, and we compare our proposed FSODS with FSODM [37] and the classical model YOLOv3 [23] and YOLOv4 [63], both of which show the superiority of the FSODS performance. Table II shows the basic information of the three datasets that we

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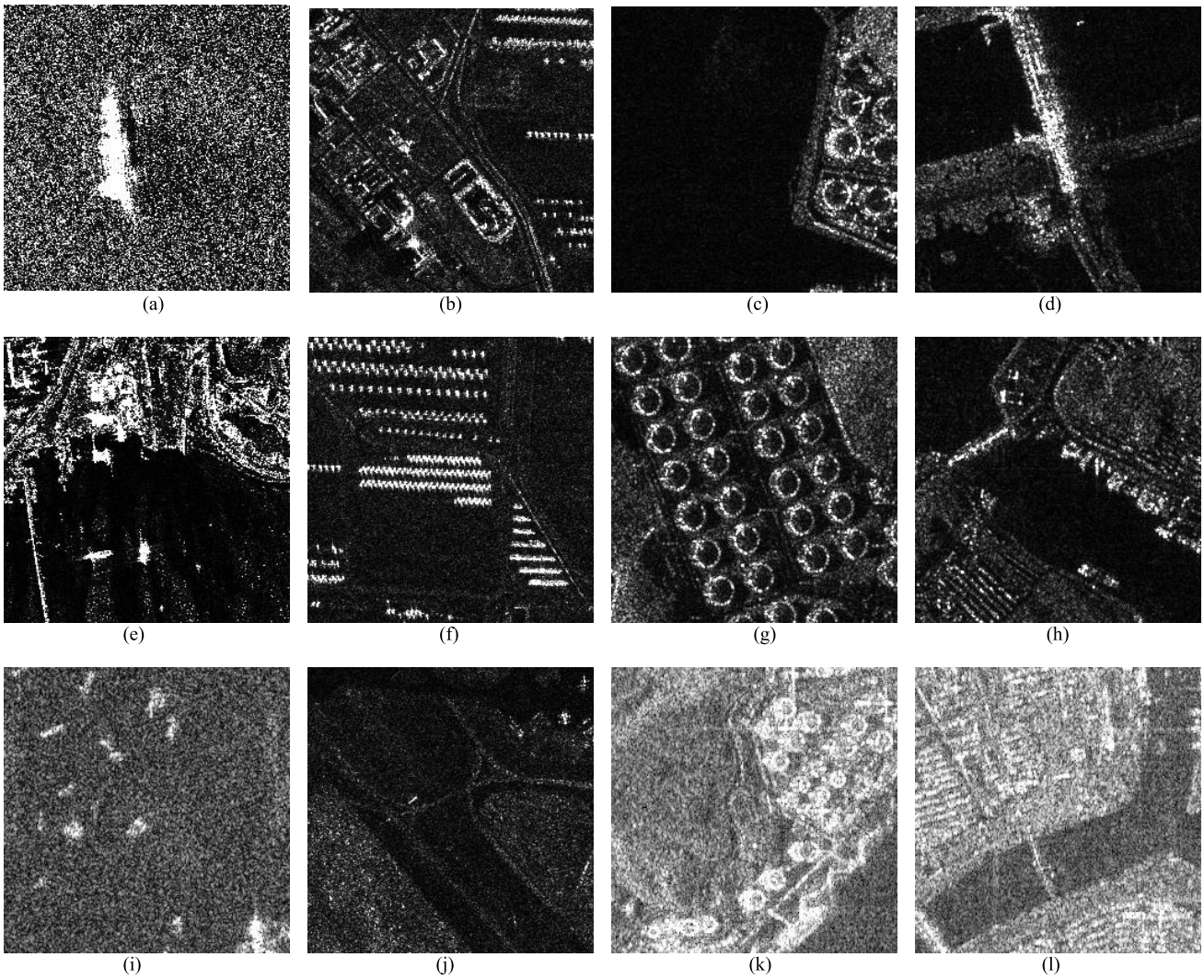


Fig. 7. Selected examples from the SMCDD-FS, which are soon-to-be-released by our research group. (a), (e), and (i) Ship. (b), (f), and (j) Airplane. (c), (g), and (k) Oil tank. (d), (h), and (l) Bridge.

used for our experiments. We will introduce them in more detail.

A. Dataset Introduction and Processing

1) SMCDD-FS: The original data of the SMCDD-FS dataset come from the soon-to-be-released SAR image multi-class target detection dataset SMCDD [19] led by our research group in the early stage. The source data of SMCDD come from HISEA-1, the first commercial remote sensing SAR satellite in China, which was developed by the 38th Research Institute of China Electronics Technology Group Corporation, Changsha Tianyi Space Science, Technology Research Institute Company Ltd. (Changsha, China), and others. Among them, our research group has used the SAR large image data captured by HISEA-1 in complex scenes to construct a large-scale SMCDD dataset for the validation of conventional target detection algorithms, which will be released in the near future [19].

Different from the original SMCDD that is produced for SAR target detection tasks under large data volume conditions, we constructed a new self-built SAR image object detection dataset for the few-shot object detection task of concern, namely, SMCDD-FS. The SMCDD-FS covers data of four target types, including ship, airplane, bridge, and oil tank. In order to facilitate training and testing, we randomly selected a part of images from the four target types of the SMCDD and constructed the training and validation sets in a ratio of approximately 7:3. The size of most images is basically 256×256, and the image size of a few bridges is 1024×1024 or 2048×2048. We will introduce it in more detail in Section IV-B. A sample of the four categories of the SMCDD-FS dataset is shown in Fig. 7.

2) SFSD: In addition to the publicly available dataset, we also validated the effectiveness of our model on the detection of close-type objects on a nonpublic SAR object detection dataset, and we constructed a new dataset called SFSD consisting of hundreds of civilian ships from the publicly available SMCDD [19] and a certain class of warships from a nonpublic dataset. The number of pictures of the warship is small, and the image size is 1024×1024; we crop it to 512×512 size and then divide it

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TABLE III
NUMBER OF TARGETS OF EACH CLASS IN THE TRAINING AND VALIDATION SETS OF THE EXTRACTED SMCDD-FS DATASET

	ship	airplane	oil-tank	bridge
Train Data	483	5238	2304	240
Valid Data	298	1130	1775	133

TABLE IV
PERFORMANCE (MAP) OF FEW-SHOT DETECTION ON THE NEW CLASS OF PART OF THE EXTRACTED SMCDD-FS DATASET

	FSODS(Ours)			FSODM[37]				YOLOv3		YOLOv4	
Class	3-shot	5-shot	10-shot	3-shot	5-shot	10-shot	20-shot	20-shot	30-shot	20-shot	30-shot
airplane	0.0182	0.0227	0.0577	0.0015	0.0020	0.008	0.0192	0.0069	0.0181	0.0346	0.0403

into a training set and a validation set according to the proportion of 8:2.

3) TFSD: In addition, we also constructed hundreds of civilian ships from the public SMCDD [19] and a few tanks and armored cars from a nonpublic dataset to form another new dataset, which we call TFSD. There are 12 large pictures including tanks and armored cars. We first divided the large picture according to the proportion of 8:4 to obtain the training set and verification set, and then crop them into 256 × 256 size, respectively, which is convenient for our training.

B. Experimental Setup

In the case of the few-shot setting, to evaluate the object detection performance of our proposed FSODS on SAR images, we randomly selected a total of 825 images from the four classes of the soon-to-public dataset SMCDD [19] and combined the number of images in each class into a training set and a validation set roughly in the ratio of 7:3. There are 572 SAR images in the training set and 253 SAR images in the validation set. The image sizes are basically 256 × 256, with a few bridges of 1024 × 1024 and 2048 × 2048. The number of objects of each class in the training set and the validation set is shown in Table III. We combine the number of SAR images and previous experiments done in the laboratory on this dataset to consider that the number of airplane objects images is the least; basically, the size on the graph is only a few pixels in size and very dense, and a 256 × 256 size graph may even have several hundred airplanes, so the performance on the benchmark is not particularly good, so, this time, the SMCDD-FS dataset selected airplane as a new class, and ship, oil tank, and bridge as the base classes for training. In addition, for the datasets SFSD and TFSD, which consist of both public and nonpublic datasets, we selected civilian ships as the base class in SFSD, civilian ships and armored cars as the base class in TFSD, warship as the new class in SFSD, and tank as the new classes in TFSD.

C. Comparing Methods

First, we compare our proposed FSODS model with FSODM [37], a model proposed in 2021 that performs well on remote sensing images, and second, similar to FSODM, we also conducted experiments on the classical object detection model YOLOv3 [23] and YOLOv4 [63]. The reason why we did not compare other methods is that there has been much previous work showing the superiority of the classical object detection algorithm YOLOv3 and YOLOv4 for standard object detection scenarios and the superiority of FSODM, which was designed for few-shot object detection scenarios on remote sensing images. The FSODS model is almost identical to the FSODM in terms of experimental parameter settings; both train a model on the same base classes dataset, then turn on new class detection on the same new class dataset, and, finally, both are validated under the same validation set. In contrast, when training the YOLOv3 and YOLOv4 models, we cull the number of targets in the new class data in the training dataset to the number that we need and then train the model normally, and the YOLOv3 and YOLOv4 models use a series of sophisticated data enhancement strategies and multiscale training methods to improve its performance, while, in the experiments to validate the FSODS performance, we did not use these training strategies. Therefore, the comparison with our method is not the fairest. The size of the input images is 256 × 256.

Our evaluation metric, similar to FSODM, is mAP, which evaluates the mAP for each class of objects, as well as the mAP in the base classes dataset and the new class dataset. mAP is calculated following the PASCAL VOC2007 benchmark [64].

D. Detection Results of FSODS on SMCDD-FS

As shown in Table IV, the performance of FSODS and the other two models for few-shot object detection on SAR images of the SMCDD-FS dataset is listed. In Tables IV and V, our FSODS model has significantly better performance on the new class of airplane than FSODM, YOLOv3, and YOLOv4 in the case that both the numbers of parameters and FLOPs are substantially reduced compared to FSODM although the mAP is not very high, which we think is due to the data, because the previous results run with the classic object detection model on the SMCDD [19], mAP is only approximately 16%, and the airplane performance was even lower; this was obtained with all airplanes involved in the training. Specifically, compared to FSODM, our mAP is 1113% higher at the 3-shot setting,

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TABLE V
COMPARISON OF THE DEGREE OF LIGHTNESS BETWEEN DIFFERENT MODELS

	FSODS(Ours)	FSODM[37]	YOLOv3	YOLOv4
Param	43.35 M	81.25 M	61.95 M	64.36M
FLOPs	2.78G	7.62G	24.74G	22.57G

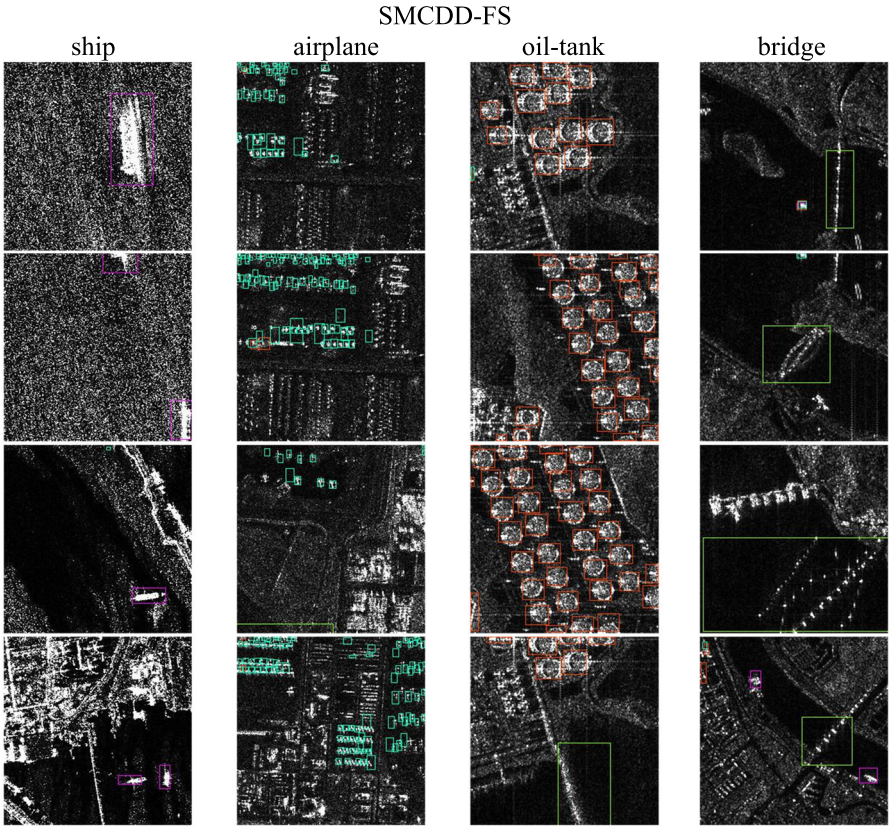


Fig. 8. Selected examples of our FSODS model results of the SMCDD-FS dataset on the new class under a 10-shot setting and on the base classes using a large number of training samples. Among them, the green box in the second column indicates few-shot object detection results on the new class, namely, airplanes.

TABLE VI
PERFORMANCE (MAP) OF FEW-SHOT DETECTION ON THE NEW CLASS OF SFSD

	FSODS(Ours)				FSODM[37]			
Class	3-shot	5-shot	10-shot	50-shot	3-shot	5-shot	10-shot	50-shot
warship	0.1364	0.2517	0.6823	0.8094	0.1450	0.2272	0.5000	0.7115

1035% higher at the 5-shot setting, and 621.2% higher at the 10-shot setting. Even the FSODM mAP at 20-shot is only 1.92%, and our FSODS performance at the 3-shot setting is similar. In contrast, the traditional object detection algorithm YOLOv3 and YOLOv4 models can only obtain 0.69% and 3.4% mAP even at the 20-shot setting and only barely match the performance of our FSODS model at the 30-shot setting.

In addition, our FSODS model is designed from a lightweight perspective, so as shown in Table V, we also counted the number of parameters and FLOPs of our model compared with FSODM. It can be seen that our FSODS has almost half the number of parameters of FSODM, 70% of YOLOv3, and smaller than that of YOLOv4. With the batch size set to 1, the FLOPs of FSODS are only 36.5% of FSODM and 11% of YOLOv3. With a much lower number of parameters and FLOPs, its detection performance for the new class still exceeds that of FSODM and the benchmark models YOLOv3 and YOLOv4.

Fig. 8 shows selected examples of our FSODS model results of the SMCDD-FS dataset in the case of the 10-shot setting.

E. Detection Results of FSODS on SFSD and TFSD

Since the warship, tank, and armored car in our test SAR images are nonpublic data, we do not show the results here, only a summary of the results. We use a part of civilian ships from SMCDD [19] as the base class on the SFSD and the warships dataset from other sources as the new class and use our FSODS and FSODM, respectively, on this dataset at the 3-, 5-, 10-, and 50-shot settings. The experimental results in Table VI show that our FSODS achieves more than 80% mAP

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TABLE VII
PERFORMANCE (MAP) OF FEW-SHOT DETECTION ON THE NEW CLASS OF TFSD

	FSODS(Ours)				FSODM[37]			
Class	3-shot	5-shot	10-shot	30-shot	3-shot	5-shot	10-shot	30-shot
tank	0.2098	0.4080	0.5493	0.4219	0.1643	0.3426	0.3937	0.4053

TABLE VIII
ABLATION STUDY OF SMCDD-FS DATASET

Meta-Feature Extractor	Aggregation Module	Focal Loss	DIoU-NMS	mAP		
				3-shot	5-shot	10-shot
Darknet53	-	-	-	0.0015	0.0020	0.0080
DarknetS	-	-	-	0.0013	0.0022	0.0213
Darknet53	AggregationS	-	-	0.0118	0.0130	0.0202
DarknetS	AggregationS	-	-	0.0120	0.0178	0.0384
DarknetS	AggregationS	√	-	0.0166	0.0205	0.0469
DarknetS	AggregationS	√	√	0.0182	0.0227	0.0577

TABLE IX
ABLATION STUDY OF SFSD

Meta-Feature Extractor	Aggregation Module	Focal Loss	DIoU-NMS	mAP
				50-shot
Darknet53	-	-	-	0.7124
DarknetS	-	-	-	0.7433
DarknetS	AggregationS	-	-	0.7941
DarknetS	AggregationS	√	-	0.8025
DarknetS	AggregationS	√	√	0.8094

on the validation set when the warships are only trained at 50 shots, while FSODM has an accuracy of approximately 71 at 50 shots, and under other shot settings, the performance of our FSODS model for the new class is basically higher than that of FSODM; thus, our model is significantly better than FSODM. We used a part of civilian ships from the SMCDD [19] and armored cars as the base classes on the TFSD and other sources of nonpublic tanks as the new class and perform a comparison test with our FSODS and FSODM at the 3-, 5-, 10-, and 30-shot settings on this dataset. The experimental results in Table VII show that the mAP of the tank as the new class in the validation set reaches more than 42% when the tank is trained at only 30 shots and is also higher than FSODM at other shot settings.

F. Ablation Experiment

We also performed ablation experiments to verify the effectiveness of our module. We performed four groups of ablation experiments on the SMCDD-FS and SFSD datasets, which are: 1) keeping the other parts unchanged and replacing the metafeature extractor with DarknetS; 2) keeping other parts unchanged, and the metafeature extractors Darknet-53 and DarknetS are added to our transformer-based aggregation module, AggregationS module; 3) keeping other parts unchanged, and replacing the loss function with the focal loss; and 4) keeping other parts unchanged, and replacing the NMS with DIoU-NMS. Tables VIII and IX show the experimental results.

The results show that our proposed lightweight metafeature extractor DarknetS outperforms the original Darknet-53 metafeature extractor, and on top of that, the performance of the model is greatly improved by our AggregationS module, especially in the case of a higher number of shots, as in the case of 50 shots in Table IX, which brings more than a five-point improvement. The focal loss and DIoU-NMS at the postprocessing end also show their effectiveness in the final results. The ablation experimental results show that the metafeature extractor, aggregation modules, the focal loss, and DIoU-NMS at the postprocessing end of our proposed method greatly improve the few-shot object detection performance of SAR images.

V. DISCUSSION

Our few-shot FSODS model not only has good performance when trained on only a few samples of a new class with labels but also should have comparable performance to traditional object detection models for base classes objects. As shown in Table X, the performance of our FSODS model on the base classes of the SMCDD-FS dataset is compared with that of FSODM and YOLOv3 and YOLOv4, which shows that FSODS performs better than FSODM on the base classes and outperforms the classical object detection model YOLOv3 and slightly lower than that of YOLOv4, which shows that FSODS

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TABLE X
PERFORMANCE (MAP) OF FEW-SHOT DETECTION ON THE BASE CLASSES OF SMCDD-FS DATASET

Class	FSODS(Ours)	FSODM[37]	YOLOv3	YOLOv4
ship	0.1682	0.2414	0.1656	0.2558
oil-tank	0.3712	0.2812	0.2970	0.3347
bridge	0.5387	0.4132	0.5428	0.5613
mean	0.3594	0.3119	0.3351	0.3839

can still achieve great performance on the base classes in a few-shot object detection problem situation in the SAR image field. FSODS even outperforms the classical YOLOv3 when the amount of data is large enough.

Second, the detection accuracy of the new class improves as the number of new class shots increases. FSODM conducted more experiments on the remote sensing dataset NWPU VHR-10 with a larger shot setting, finally showing that only 60 (8%) training samples were from the new class, and the detection performance could be almost identical to the baseline model using all training samples. However, it also depends on the detection difficulty of the specific dataset, as in our SMCDD-FS dataset. When the number of airplanes in the new class was increased to 150 shots, our FSODS was able to achieve an mAP of 9.09% on the new class, which is not included in Table IV. Although it did not reach the accuracy of the baseline model without the few-shot setting, this is due to the data itself, and 150 shots are only 2.8% of the number of shots in 5238 of the original training set, which is close to the accuracy of the benchmark dataset.

Most of the existing work has resorted to classical mature methods in the field of traditional object detection, such as SSD and Faster-RCNN, using a two-stage approach to complete the detection, making the inference after model deployment slow and mostly unable to meet real-time requirements. Second, although the candidate box-based detection method is effective, it also increases the complexity of the model structure, leading to a higher demand for device memory in the model deployment phase. Therefore, especially in the military field, the lightweight design of few-shot object detection is particularly important so that our airplane can easily carry embedded devices with ported few-shot object detection models and can make accurate judgments more quickly.

VI. CONCLUSION

This article introduces a lightweight object detection algorithm applied in the military field that can effectively implement few-shot object detection of SAR images through our transformer aggregation module. This is a preliminary examination of few-shot object detection in the field of SAR images, and it is also a problem that needs to be solved in the military field. In this article, inspired by the FSODM model for remote sensing images, we design and improve its lightweightness and adapt it to the characteristics of SAR images. Then, to better learn the important features of objects in SAR images and the interclass feature variations, and weaken the irrelevant noise features, we propose an effective aggregation module AggregationS and alleviate the sample imbalance and the postprocessing anchor frame suppression caused by sample imbalance to improve the accuracy of few-shot detection. We design a focal loss that is more adapted to sample imbalance training and a DIoU-NMS at the postprocessing end. To verify the effectiveness of the experiments, we constructed several self-built benchmark datasets for few-shot object detection of SAR images in real scenes. Finally, the experiments on the self-built publicly available SAR image object detection dataset SMCDD-FS and two self-built nonpublic SAR image datasets SFSD and TFSD show that the detection accuracy of our proposed method on only a small number of labeled samples exceeds that of FSODM and the classical object detection algorithm YOLOv3 and YOLOv4, and the numbers of parameters and FLOPs are greatly reduced. This work is now a further exploration of few-shot learning for SAR image classification, evolving the classification task to the object detection task, which is also a further attempt to perform few-shot object detection on SAR images. In the future, we will further explore more effective few-shot object detection methods in the field of SAR images and try to combine the inherent physical mechanisms of interpretable SAR images and learnable few-shot object detection algorithms based on deep neural networks for complementary advantages, which have a wide range of development prospects.

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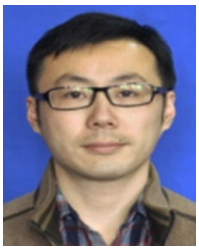
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